.Scora

Trust Skills, Not Portfolios.

Scora - AI Dependency & Developer Verification Research

Evidence behind Scora's understanding-first verification model

Research folder

Product: Scora **Team: Code Luck**
Phase: Product Validation & Strategy **Date: 27 July 2026**

Research question

How can a freelance platform distinguish a developer who responsibly uses AI while retaining engineering ownership from an applicant who can generate plausible code but cannot explain, debug, secure, modify, or maintain it?

Corrected product thesis

What Scora verifies

Scora does not attempt to prove that code was typed without AI. It verifies that the person claiming the skill can take engineering responsibility for the result. A candidate who only produces output but fails explanation, debugging, security review, or modification gates does not receive verification for that skill track.

Evidence synthesis

Finding	Evidence	Meaning for Scora
AI output can look correct while remaining unreliable	Stack Overflow 2025: 66% cite almost-right AI answers; 45% cite extra debugging time.	The assessment must include hidden faults and debugging, not only generation.
Confidence can exceed security performance	Perry et al. observed less-secure solutions and higher perceived security in its AI-assisted group.	Candidates must identify threats and negative cases; self-confidence is not evidence.
AI can improve experienced work	GitHub's RCT with 202 experienced developers reports better functionality and modest quality gains.	Do not ban AI users. Verify ownership and engineering judgment.
Effects vary by task and context	USENIX Lost at C found only a small security difference in its tested C task; METR found a slowdown in familiar-repository work.	No universal AI penalty is justified. Scora measures the candidate on role-relevant tasks.
Hiring signals are under pressure	A study of 32 industry professionals reports GenAI complicates evaluation of true candidate ability.	The interview must be grounded in the submitted code and include live changes.
Independent review is required	OWASP recommends full-diff review, independent tests, dependency verification, sandboxing, and human accountability.	Scora combines deterministic checks, code review, interview evidence, and selective human review.

The failure Scora is designed to catch

- Output without a mental model: the candidate cannot describe data flow, state, dependencies, or failure modes.
- Copy without review: the candidate accepts generated code, packages, tests, or configuration without independent verification.
- Success on the happy path only: the solution passes visible tests but fails boundaries, invalid inputs, concurrency, authorization, or recovery cases.
- No change ownership: the candidate cannot implement a small requirement change without regenerating the entire solution.
- No production responsibility: the candidate cannot debug logs, explain trade-offs, assess security, or plan maintenance.

Assessment design derived from the evidence

Stage	What it measures	Failure signal
1. Fundamentals checkpoint	Core language, debugging, data flow, tests; AI restricted for this stage.	Cannot reason about basic behavior without generation support.
2. AI-allowed engineering task	Ability to direct AI, review output, choose architecture, and deliver working code.	Blind acceptance, weak decomposition, unsafe dependencies, or missing tests.
3. Code-grounded interview	Explanation of the candidate's own submission, assumptions, and trade-offs.	Answers do not match the code or remain generic.
4. Hidden-bug diagnosis	Ability to locate, explain, and repair a defect.	Cannot trace behavior or validate the fix.
5. Live requirement change	Ownership, adaptability, regression awareness, and maintainability.	Regenerates blindly or breaks existing behavior.
6. Evidence decision	Combined correctness, quality, understanding, security, integrity, and confidence.	Required understanding gate not met: track remains unverified.

Competitor and market gap

Product category	Representative examples	Core signal	Scora opportunity
Freelance marketplace	Upwork, Fiverr	History, satisfaction, reviews, portfolio, badges, identity.	Verify current engineering ownership before placement in a trusted developer pool.
Assessment platform	HackerRank, Codility	Tests, real-world tasks, interviews, AI policies, integrity controls.	Make verified evidence persistent, client-readable, and connected to freelance discovery.
Curated network	Toptal and agencies	High-touch screening and admission.	Expose evidence, confidence, track scope, recency, appeal, and reassessment.
Developer evidence	GitHub, portfolios, certificates	Artifacts, activity, claims, and completion.	Test whether the claimant can explain, change, debug, secure, and maintain the artifact.

Claims that still require Scora's own validation

Hypothesis	Test	Decision evidence
Understanding gates reduce false positives	Blind senior reviewers rate 50 submissions before and after interview/change evidence.	Material number of test-passing candidates fail ownership evidence.
Responsible AI users are not unfairly rejected	Compare calibrated outcomes across declared AI use and no-AI paths.	Equivalent engineering evidence produces equivalent verification decisions.
Clients act on ownership evidence	Profile prototype test with SME buyers.	Evidence changes shortlist order and reduces requested screening time.
The gate is explainable and appealable	Moderated report and policy test with developers.	Users can explain why they passed/failed and how to challenge an error.

References

We selected sources for their direct relevance and authority. Company pages describe current product behavior; surveys and research papers support market and risk claims. Assumptions that still require interviews or field testing are identified in the validation sections.

1. Stack Overflow 2025 Developer Survey - AI. Among respondents answering the accuracy question, 46% distrust AI-tool accuracy versus 33% who trust it. The most reported frustration is output that is almost right, while 45% cite extra time debugging AI-generated code.
https://survey.stackoverflow.co/2025/ai

2. GitHub randomized controlled study on Copilot and code quality. A study with 202 developers who had at least five years of experience found improved functionality and modest quality gains with Copilot. AI use alone is therefore not evidence of low skill.
https://github.blog/news-insights/research/does-github-copilot-improve-code-quality-heres-what-the-data-says/

3. Do Users Write More Insecure Code with AI Assistants?. In a security-focused user study, participants with an AI assistant produced less secure code and were more likely to believe their code was secure; lower trust and more active engagement correlated with fewer vulnerabilities.
https://arxiv.org/abs/2211.03622

4. Lost at C - USENIX Security 2023. A different user study found only a small security impact in its specific C task. This mixed evidence shows that risk depends on the task, user, and verification process rather than AI use alone.
https://www.usenix.org/conference/usenixsecurity23/presentation/sandoval

5. METR - Early-2025 AI and experienced developer productivity. A randomized study of experienced open-source developers working in familiar repositories found early-2025 AI tools slowed completion in that setting, despite participants expecting a speedup.
https://metr.org/blog/2025-07-10-early-2025-ai-experienced-os-dev-study/

6. Generative AI code tools and software-engineer hiring. A study of 32 industry professionals reports that GenAI complicates evaluation of candidates' true abilities, while many participants value candidates who can demonstrate effective and informed AI use.
https://arxiv.org/abs/2409.00875

7. OWASP Secure Coding with AI. OWASP recommends independent verification, full-diff review, adversarial testing, dependency checks, human accountability, and sandboxed execution for AI-assisted development.
https://cheatsheetseries.owasp.org/cheatsheets/Secure_Coding_with_AI_Cheat_Sheet.html

8. Upwork Job Success Score. Upwork's Job Success Score reflects client satisfaction, contract outcomes, and long-term client relationships. It is a reputation signal rather than a direct test of code ownership or current engineering understanding.
https://support.upwork.com/hc/en-us/articles/38437362753171-What-is-a-Job-Success-Score

9. Upwork - Build a complete profile. Upwork emphasizes work history, certifications, availability, achievements, keywords, and portfolios as profile trust inputs.
https://support.upwork.com/hc/en-us/articles/34924882793107-Build-a-100-complete-profile

10. HackerRank Screen - AI-era skills and integrity. HackerRank separates fundamentals without AI, ability to work with AI, and AI concepts, while adding identity and assessment-integrity controls. This validates multi-mode assessment in the AI era.
https://www.hackerrank.com/products/screen

11. Codility Task Library. Codility combines real-life skills, problem solving, and technical knowledge tasks, supporting a rounded assessment rather than a single puzzle score.
https://support.codility.com/hc/en-us/articles/360043827713-The-Codility-Task-Library

12. Docker Engine security. Docker documents namespaces, cgroups, capabilities, daemon attack surface, and hardening controls relevant to isolated code execution.
https://docs.docker.com/engine/security/